

Applying GANs for Image Synthesis and Recognition in Forensic Contexts

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Abstract—Forensic investigations usually depend on manually drawn facial sketches that are based on witness descriptions, which can result into inaccuracies and time delays. This paper proposes a method of automating face sketch generation using Generative Adversarial Networks (GANs), specifically StackGAN, through deep learning. The system uses Natural Language Processing (NLP) techniques to extract relevant facial features from witness descriptions before creating high-resolution facial sketches using a two-phase StackGAN model. To enhance precision, sketch generation is matched with a criminal database using Convolutional Neural Networks (CNNs) for suspect identification. The proposed method was tested on well-known public datasets like FFHQ and CelebA, showing noticeable improvements in how realistic the sketches looked and how accurately they resembled real faces. This approach can make forensic investigations more efficient by cutting down on human error, speeding up the process of identifying suspects, and offering a scalable, AI-powered tool for law enforcement.

Keywords—Text-to-Sketch Generation, Convolutional Neural Networks, Forensic Face Recognition, Image Synthesis, Criminal Identification, Generative Adversarial Networks StackGAN

I. INTRODUCTION

Through forensic sketching law enforcement has maintained a crucial investigation tool for centuries which helps create images of suspects starting from eyewitness testimonies. Traditional forensic sketching requires skilled forensic artists for manual drawing that depends heavily on witness memory and artistry quality which can affect the accuracy of suspect identification [2]. The method of creating forensic sketches through human artists has many drawbacks such as delayed procedures and unreliable results that reduce both investigation efficiency and suspect identification rates. Errors resulting from human memory along with use of artistic interpretation negatively affect forensic investigations, as discussed in [4].

Emerging developments in artificial intelligence and deep

learning open new avenues for automation in sketch generation. GANs have emerged as a highly effective technology for image generation, with models like StackGAN achieving realistic results [13]. The system uses two networks where the generator produces images while the discriminator measures how closely generated images align with actual real-world data. The adversarial framework of GAN development enhances the system's ability to learn complex facial patterns which allows it to create highly realistic images thus making them suitable for forensic sketch work [1].

First of all, this system processes the eyewitness description through Natural Language Processing (NLP) techniques that decide which features in a facial description of an eyewitness include hair color, structure, facial features, and some distinguishing ones.[4] These features are then mapped onto a StackGAN architecture that is intended to progressively refine the image and transform coarse sketches into high-resolution and detailed representations of the suspect's face. The consistent and accurate outputs of this approach are ensured by a comprehensive preprocessing pipeline for facial data, including normalization and augmentation, which enhance the diversity of training samples and improve model generalization across descriptions. The generated facial sketch is then passed through a convolutional neural network (CNN) for suspect identification. That is, the sketch is compared to an image database. A CNN draws embeddings from both the generated sketch and an image database. These embeddings then have similarity metrics applied to find potential matches. Therefore, GAN-generated sketches coupled with a facial recognition system provide an end-to-end solution for both sketch generation and suspect identification. As in [1], the Tunable Latent GAN (TL-GAN) is further used to fine-tune the output to allow specific control over some facial features based on the description fed to the system.

Being better than the traditional manual sketching practices, the system proposed here would make it a timely, accurate, and scalable solution for the investigation. The subsequent sections

present the background literature related to GANs, or Generative Adversarial Networks, and forensic applications, a detailed discussion of the proposed model, and so on. Then it presents some experimental results. Finally, the paper concludes with an insight as to how this area might develop further in the future.

II. LITERATURE REVIEW

In order to enhance suspect identification, Jalan et al. [1] developed a system that leverages Generative Adversarial Networks (GANs), specifically StyleGAN, to improve suspect identification. Their approach generates facial images based on input descriptions and further enhances them using Tunable Latent GAN (TL-GAN), which adjusts the input in the latent space. By reducing user input and accelerating the process, their approach greatly enhances identification accuracy, rendering it a useful tool for real-world crime investigations. In another study, Patil and Shubhangi [2] addressed face recognition by forensic sketch through a geometrical face model. They employed an AdaBoost approach for face detection and added a geometrical model to extract discriminative facial features. An additional artificial neural network (ANN) classifier enhances precision. Although impressive results are delivered, the work points out how improved feature extraction techniques are desired to deliver still better results for forensic purposes.

Dixit and Raj [3] went the more user-centric route by creating a system with a drag-and-drop interface through which users can simply draw facial sketches. These are then compared to a criminal database with the help of the DeepFace facial recognition library. The research highlights the need to improve this process further to increase the accuracy of recognition. Jha et al. [4] combined CNNs and GANs as a hybrid model to recognize face sketches. The CNNs assist in feature extraction from forensic sketches, and the GANs produce realistic images to identify them. Training the system on a big database of sketches and real photographs significantly enhances its robustness and accuracy of sketch-to-photograph matching [4]. Wu et al. [5] proposed a new sketch-to-sketch transformation model based on conditional GANs (cGANs). The method iteratively improves early forensic sketches from ambiguous or partial descriptions to assist forensic artists in creating more precise and detailed representations, leading to greater chances of successful identification.

Building on facial sketch recognition, Reed et al. [6] investigated how age progression can influence sketch identification. Using a recurrent neural network (RNN), their system keeps up with changing facial features with age, thus it can recognize aged sketches and real-time photographs. This methodology manages to cope effectively with the vicissitudes of natural aging and is, therefore, a useful resource in the long-term prosecution of crime [6]. AttnGAN, introduced by Xu et al. [7], employs an attention mechanism to produce very detailed images from textual descriptions by highlighting the most pertinent words for each sub-region of an image. It also employs a deep multimodal similarity model to enhance the precision of matching an image to textual descriptions. Showing a breakthrough improvement on the CUB and COCO datasets, this model represents a breakthrough in text-to-image

synthesis. Ouyang et al. [8] addressed the challenging problem of cross-modal face matching, especially in the case of forensic sketches and caricature sketches that tend to involve abstraction and exaggeration. To this, they introduced a mid-level facial attribute model that learns semantic properties separately for each modality and is thus robust to distortion and misalignment. They also merged these qualities with novel characteristics through Canonical Correlation Analysis (CCA) to result in robust outcomes. They also provided a new dataset with 59,000 attribute labels to enable future studies.

Thakare et al. [9] suggested a CNN-based method to identify forensic face sketches using both global and local features with a fusion model. This approach improves accuracy in recognition and provides a stable solution to the difficulties of law enforcement agencies while identifying suspects. Galea and Farrugia [10] tackled the challenges of forensic sketch-mugshot gallery matching using a deep learning method optimized for small data. By transferring a pre-trained face recognition model through transfer learning and employing a 3D morphable model to generate synthetic images and sketches, they enhanced accuracy and minimized error rates. This novel framework has potential for enhancing criminal identification via deep learning methods. Zhao et al. [11] addressed the problem of generating photorealistic facial images from sketches, thus confronting this challenge as face hallucination. This GAN-based model uses both the outline and attribute vectors to better contribute to an image that attains realistic completeness with respect to outlines and attributes. The generator part of the network includes a feature extraction and downsampling-upsampling layer connected through skip connections to maintain performance with fewer layers while ensuring that the generated face matches the desired attributes through the discriminator. The two-branch structure of the network enables it to capture not only the low-level profile of the object but also the high-level semantic details, thus providing highly accurate results compared to other image translation methods.

The research study by Kumar et al. [12] discussed generating face images driven by GAN from text descriptions. Generally, there used to be much talk in such fields about birds or flowers, for example. A two-stage StackGAN architecture is applied for a sketch-refinement scheme, which interprets particular attributes of the face, like hair color and expressions, to produce realistic images, opening up multiple applications in fields ranging from art and entertainment to education. Their model achieves an inception score of 4.04 on the CelebA dataset [21], indicating that it can indeed be used for translating complex textual descriptions into high-resolution realistic images of faces [12].

Zhang et al. [13] presented the problem of generating high-quality images with GANs. The authors introduce StackGAN, a two-stage approach in which Stage I generates a low-resolution image using a text description, while Stage II refines that image into a high-resolution photorealistic output. Additionally, StackGAN-v2 introduces a multi-stage architecture with multiple generators and discriminators, improving stability and image quality. Some proposed models have reportedly outperformed existing methods in realistic

image synthesis and are considered significant advancements in the realm of text-to-image synthesis.

Ayanthi [14] examined the task of generating photorealistic facial images from text descriptions using StyleGAN2. The developed framework uses BERT embeddings, which map text to the latent space of StyleGAN2, enabling the production of aligned facial images at a resolution of 1024x1024 pixels corresponding to input descriptions. This model performs better than earlier approaches, achieving a 57% similarity to ground truth images and a face semantic distance of 0.92. Promising results are achieved in both image quality and alignment stages.

Ramzan et al. [15] raised the challenge of generating realistic images from text descriptions in their research. Based on this work, a Recurrent Convolutional Generative Adversarial Network (RC-GAN) has been proposed that can generate semantically consistent images by converting visual concepts into pixels. Trained on the Oxford-102 flowers dataset, their model achieves an inception score of 4.15 and a PSNR value of 30.12 dB, demonstrating its ability to generate realistic flower images from captions. Future work may involve expanding this model into multiple datasets for wider applications.

Martis et al. [16] produced human facial sketches from textual descriptions with the help of the Generative Adversarial Network. The study aims to fill the gaps in sketches provided by an eyewitness. The authors suggest the use of an Inverse Gamma Correction Technique during training to make the sketches more vivid. This method, considered an improvement over existing techniques, achieves an inception score of 1.438 ± 0.049 and a Frechet Inception Distance of 65.29. Klum et al. [17] discussed the effectiveness of forensic and composite sketches in automated facial sketch recognition for law enforcement. The authors present a special-purpose mugshot and sketch database, comparing holistic and component-based methods for sketch- to-photo matching. Results show that composite sketches are more likely to yield positive matches to mugshots compared with forensic sketches, especially in an age and gender and race or ethnicity demographic filter, and thus represent promise for automated systems in supplementing more traditional means of suspect identification. Vidhyap [18] introduced a deep learning model using DCGAN (Deep Convolutional Generative Adversarial Network) for the conversion of low-quality forensic sketches into high-resolution realistic images. Prepared sketches are used in the training process to formulate a real and accurate facial image over various epochs. Forensic sketch-based suspect identification would be improved in quality as well as realism to enrich accuracy in face recognition applications in security and entertainment. Bushra and Maheswari [19] presented a technique utilizing Deep Convolutional Generative Adversarial Network (DCGAN) for transforming forensic sketches into high-resolution facial images. This method addresses the challenge of identifying suspects based on rough, hand-drawn sketches created from eyewitness descriptions. By training the neural network on these sketches, the model progressively generates accurate, realistic images of the suspect, enhancing the speed and precision of crime investigations. The system proves especially useful in forensics, law enforcement, facial recognition, and security applications, enabling quick

identification from incomplete or low-quality sketches with variations in pose, color, and tone.

The reviewed literature offer diverse approaches to improving face sketch generation and recognition systems. By employing advanced technologies like GANs, CNNs, RNNs, and facial recognition libraries, each study contributes to refining the accuracy, speed, and user-friendliness of forensic tools. Collectively, they highlight the need for ongoing research into integrating these techniques, ultimately enhancing the efficiency and effectiveness of crime-solving efforts in real-world applications.

III. PROPOSED MODEL

Our proposed system employs an advanced GAN and facial recognition technique in our methodology for forensic face sketching and identification. We ensure the accurate and efficient generation and comparison of forensic sketches with these stages divided into critical steps starting from the selection of preliminary data, feature extraction, to a detailed evaluation of system as shown in Figure 1 and Figure 2.

For our project on forensic face sketching and identification, the basis of our data collection methodology will be high-quality facial images carefully acquired to ensure a range of diverse and representative data that can support model training and evaluation. The primary sources of our data are the widely recognized benchmarks in the research community: [12] the datasets are the CelebA and FFHQ, which is the Flickr-Faces-HQ dataset. These datasets include facial images of various demographics and thus form a good foundation for our training on GAN models to generate an accurate forensic sketch [21][22]. We also consider other contextual information that would be vital in enriching our data. Essential demographic information such as age, gender, and ethnicity are considered to make sure that the model captures facial representations that are comprehensive for diverse population groups. This added context aims at enhancing the possibility of the model generating and matching realistic facial sketches for all possible individuals. Besides, we integrate facial images from other sources available for us, such as publicly accessible datasets and collaborations with academic institutions. Thus, a multi-source approach will result in exposure of our model to a broader spectrum of facial variations, hence the increased ability to learn complex patterns for accurate forensic identification.

A. Data Pre-processing

The pre-processing of the data is important so that the GAN may actually generate facial images conditioned on good-quality text. It will include cleaning and standardization of text data as shown in Figure 3. Raw textual descriptions will first be tokenized followed by the removal of unnecessary symbols or irregularities. The use of POS tagging and entity recognition is first applied to obtain key descriptive attributes based on "eye color" or "face shape." The descriptive text is next converted into numerical vectors using a word embedding technique, such as Word2Vec or the LSTM-based model [7]. These vectors then serve as conditioning inputs for the GAN. The embedding captures the semantic information in the descriptions to help the model accurately interpret a complicated description.

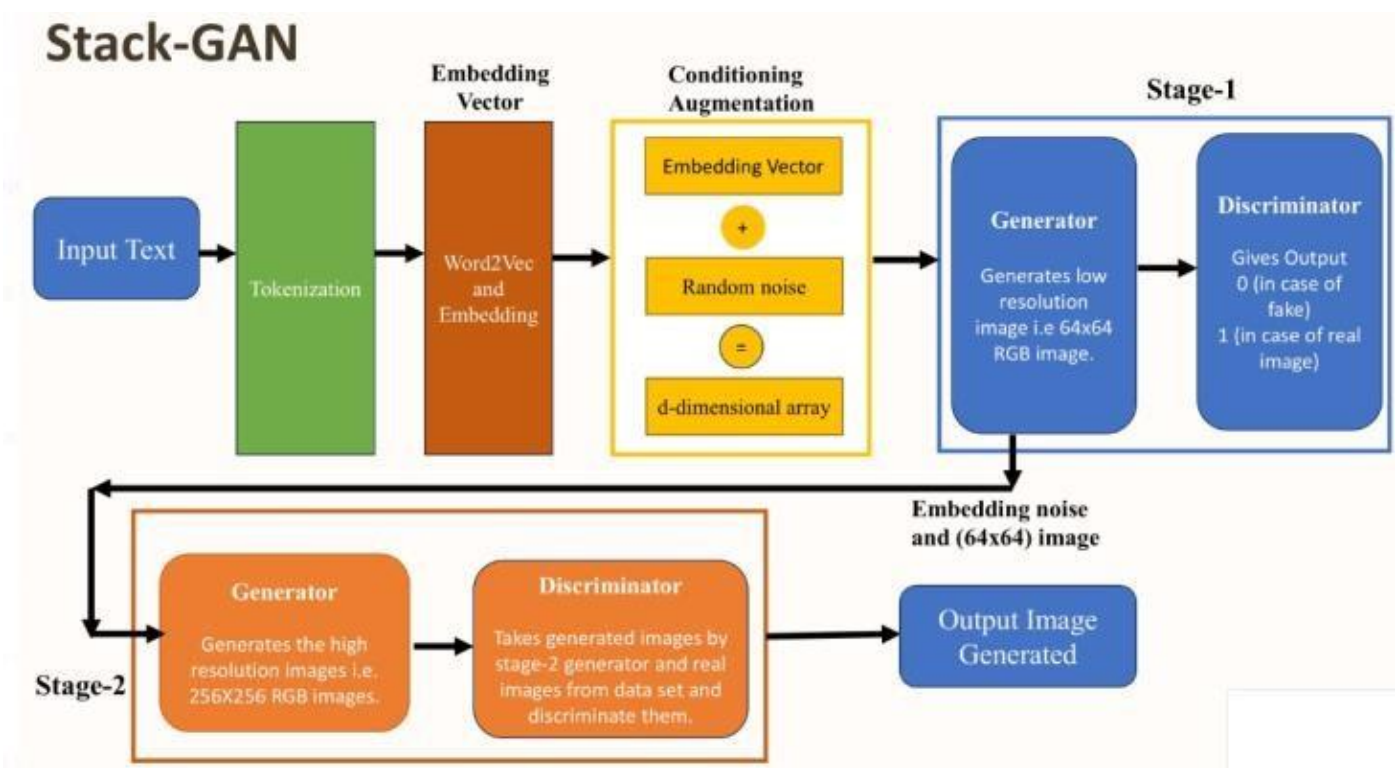


Fig. 1. Diagrammatic View of Proposed Model

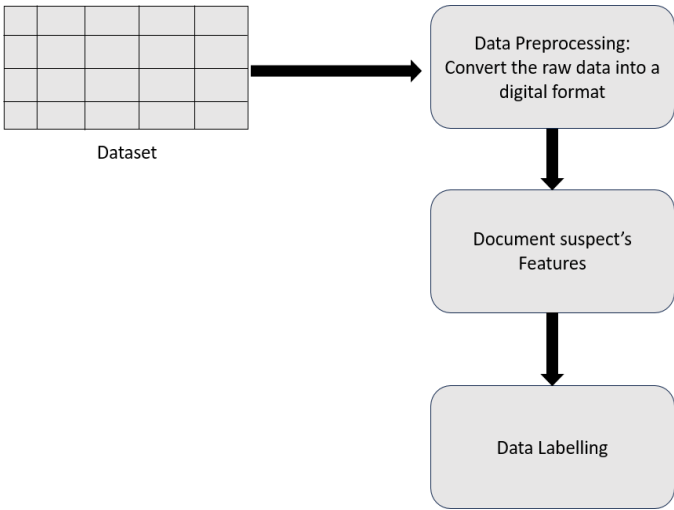


Fig. 2. Data Collection Process

B. Data Collection

For image data, any real images used in training are resized and normalized to have consistent resolution, usually in 256x256 dimensions, corresponding to that architecture of the GAN as shown in Fig.2. It eliminates backgrounds or normalizes them to highlight the facial features rather than interrupting unwanted noises while the generation process goes through. Attribute-annotated datasets like the CelebA were aligned and cropped facial features consistently for uniform input to enhance the ability of GANs in linking the features of the text with those of images [21][22]. Pre-processing pipeline helps reduce the difference between the descriptive text and the

generated images, such that producing reasonable output and exact details will be improved.

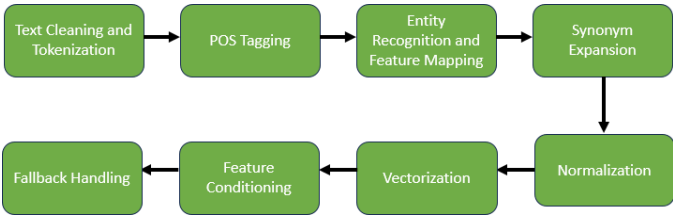


Fig. 3. Data Preprocessing

C. Generative Adversarial Network (StackGAN)

Imagine a Generative Adversarial Network, or GAN, as a kind of two-player game where two "players" — the generator and the discriminator — are constantly learning from each other. The generator's job is to create realistic-looking data, such as images, starting from random noise. Meanwhile, the discriminator is trained to tell the difference between real images (from a dataset) and those the generator creates. As the game progresses, the generator learns to make its fake images more and more convincing, while the discriminator sharpens its ability to detect what's real and what's fake, as discussed in [12]. Through this ongoing back-and-forth, both parts get better at their tasks, with the generator eventually producing images that are nearly indistinguishable from real ones.

Training a GAN involves a minimax game, where the generator tries to "fool" the discriminator as often as possible, and the discriminator tries to become better at spotting fake images. This cycle continues until the generator can produce highly realistic images that are difficult to tell apart from real

examples as shown in Fig.4. This process is dynamic and iterative — each side improves with feedback from the other. As a result, GANs are capable of creating high-quality synthetic data, which can be used for a variety of applications, like generating images, transferring styles, or enhancing image resolution.

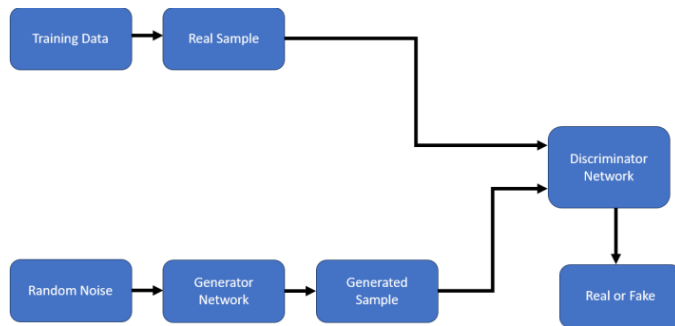


Fig. 4. Working of GAN Model

StackGAN builds on the original GAN concept to produce high- resolution images through a clever two-step process. In the first step, it starts with a rough, low-resolution image generated from a textual description, capturing only the basic layout and shapes from the description. StackGAN completes its second stage to add the finer details and enhance the first coarse sketch to a clear realistic image, sketch to a distinct, realistic image. The secondary step allows the generator to first create a sketch before it adds textures and facial features to the image. StackGAN creates very detailed real-like images by implementing a two-step process which improves the quality beyond what typical GAN-generated images can achieve where they are prone to not attaining high resolution vision in one process [13].

A two-stage system by StackGAN aids forensic face sketching activities effectively. Witness testimony allows StackGAN to generate clear facial sketches of high resolution. The sketch accurately mirrors the descriptive features of the witness because of the two-stage refinement mechanism. The technology minimizes human interaction, accelerates the process of generating sketches, and offers law enforcement an accurate tool to help identify suspects. Application of StackGAN in forensic investigations is a giant leap forward that illustrates how AI can assist in enhancing accuracy and efficiency in such important areas of law enforcement.

D. Face Recognition

In the second phase of the project, following the generation of the forensic facial sketch in Phase 1, onward with more critical processes for recognition of the face. First is the preprocessing of the generated image, to align and resize the face image. This is an important step because clustering the core parts of the facial features such as the eyes, nose, and mouth on the same horizontal line acts as a standard reference. By aligning these landmarks, a greater degree of image-rigidity is achieved, where distortions caused by variation in angle, lighting, or resolution are corrected; the image is now ready for the next stage of feature extraction and comparison.

The next step is still Facial Feature Extraction, which is done through the Convolutional Neural Network (CNN) [7],

distinguishing the individual properties of the face from the preprocessed image. Afterward, these features get converted to a compact form known as feature vector that highlights the major attributes of the individual face. In the next step of database matching, the feature vectors are now compared against a criminal database. At this stage, the Cosine Similarity method is applied, comparing the angle between the feature vectors to find the closest match between the generated facial sketch and the records stored in the database.

To this end, we also built up on this approach by introducing confidence scoring and thresholding techniques based on Siamese Networks. The Siamese network gives a score assigned to how closely the produced facial sketch copies that of already stored images in the database and, on the basis of this score, whichever the guess might happen to be, is either taken as match or rejection. If a score couldn't cross the threshold that means there's no match, suggesting that the generated sketch doesn't have a corresponding person in the database. This phase is systemically crucial because it rectifies the assurance of the recognition process, finally providing a conclusive identification of the author of the draw, and concluding the forensic process.

IV. IMPLEMENTATION AND RESULTS

The proposed system was trained and evaluated on FFHQ (Flickr- Faces-HQ) and CelebA [21] (CelebFaces Attributes Dataset) dataset. FFHQ [22] provided facial images of high quality and diversity, which ensured that the forensic sketches captured precise details across different demographics. Annotated facial attributes in Celeb A made it possible for the model to learn facial feature correlations effectively. The main difficulty occurred when we modeled the training to include specific demographic data while showing reduced performance for populations that remain underrepresented.

The main limitation of Western-dominated training is that models often perform poorly on the creation and recognition of faces belonging to Asian and African ethnicities. Such biases could result in fairness issues when performing forensic investigations using face sketches. The dataset contains both FFHQ and CelebA collections to enhance diversity through variation of facial characteristics across different gender groups and age ranges and ethnic backgrounds.

Additionally, to prevent overfitting, techniques such as data augmentation (rotation, flipping, and contrast adjustments) and dropout regularization are employed. Future work will focus on bias detection techniques and fairness- aware GAN models to improve equitable performance across all demographics [21] [22].

Preprocessing steps included text cleaning, tokenization, and embedding conversion using Word2Vec for witness descriptions, while image normalization ensured uniform 256×256 resolution for all dataset images. To improve model generalization, data augmentation techniques like flipping, rotation, and contrast adjustments were used. Sketch generation was accomplished using a two-stage process using the StackGAN-based architecture. Stage I generated an initial low-resolution sketch, while Stage II refined the details, improving image clarity and realism. The model was trained for 200

epochs using the Adam optimizer with a learning rate of 0.0002.

Experimental results from previous studies have demonstrated significant improvements in GAN-based face synthesis and identification [20]. A study analyzing performance across multiple architectures reported the results as shown in the Table I. These results validate the feasibility of the proposed system for enhancing forensic investigations through automated AI-driven sketch generation and suspect identification.

TABLE I. SSIM PERFORMANCE OF GAN-BASED FACE SYNTHESIS MODELS

Model	SSIM Performance		Datasets
	Photo Synthesis	Sketch Synthesis	
Standard GANs	0.5517	-	CUFSF
DCGANs	0.587	-	CUFSF
Multi-Adversarial Networks	0.7915	0.6156	CUFS, AR, XM2VTS
GFP-GANs	0.78	-	CUFS, CUFSF, XM2VTS, AR
STF CycleGAN	0.637	-	CUFS, CUFSF, XM2VTS, AR
STF CycleGAN + GFP-GANs	0.8154	-	CUFS, CUFSF, XM2VTS, AR

V. CONCLUSION AND FUTURE SCOPE

This study provides an excellent solution for forensic experts, particularly in instances with no photo evidence. With the application of GAN and StackGAN, it produces realistic facial sketches descriptions, filling an important in forensic identification. The model also further improves recognition by considering age, gender, and ethnicity through a diverse dataset. By improving deep learning methods, it supports investigations in the identification of suspects or missing individuals. This study highlights the capabilities of AI-based tools in law enforcement, rendering forensic sketching faster and more accessible. Future research involves enlarging the dataset, improving deep learning models, and adding explainable AI. Real-time sketch-to-photo translation and cooperation with police enhance real-world application and identification rate.

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